Ajay Venkatesh

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Skills

Programming Languages: HTML, CSS, TypeScript, JavaScript, Python, Java, (ES6)
Web Technologies: REST API, GraphQL API, Microservices, React, Node.js, Express.js, Sequelize
Database: RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse
Cloud: AWS (AWS CDK, EC2, S3, API Gateway, Lambda, SageMaker), Azure (OpenAI services, Function Apps, Storage blobs)
Tools: Git, Visual Studio, Cursor, Jupyter, Postman, Kafka, Kubernetes, Docker, Power BI
AI/ML: PyTorch, Hugging Face, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, SVM, Random Forest, MLflow, PySpark

Experience

Campus Mesh **Jan 2026 – Present**
Software Engineer *New York, US*
- Collaborated and maintained scalable **Node.js microservices** powering real-time academic workflows for **10K+ users**, including study groups, messaging, notifications, and profile management.
- Built a personalized **recommendation engine** using collaborative filtering and behavioral analytics, increasing user engagement by **28%** and retention by **50%**.
- Authored **LLM-based retrieval pipelines** for CampusIQ (AI Assistant), improving contextual accuracy by **30%** through semantic retrieval and structured data indexing.

Amazon **May 2025 – Aug 2025**
Software Development Engineer Intern *Seattle, US*
- Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the **system design** for approval and established Role Based Access **(4+ roles)** using **OAuth, Midway SSO, and HELIS**.
- Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by **70%** and enabling **multi-stage, multi-region CI/CD deployments**.
- Built a **React 18** frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by **40%**.
- Engineered backend services in **Java** using **Coral, Smithy, Dagger**, and **CBOR serialization**; applied low-level design and design patterns, reduced backend execution time by **35%** and ensured reliability by unit and integration testing using **Mockito**.

New York University (Novel AI Technologies, Inc.) **Aug 2024 – Present**
Software Developer *New York, US*
- Architected an **NSF-funded ($100K)** AI-powered health app for **iOS** and **Android**, processing real-time wearable data; deployed cloud-hosted AI models to analyze health metrics and process food images.
- Engineered ETL pipelines transforming **NoSQL** into **PostgreSQL** using **concurrency control and locking mechanisms**; processed **250K+** records and powered **AWS QuickSight** dashboards for real-time health analytics.
- Developed an insurance analytics platform **(React, Node.js, REST APIs)**; set up **Docker** containers on **EC2**, implemented Stripe billing, RBAC, and row-level security, generating **$50K+** revenue.
- Automated underwriting via broker portal integrated with insurer platform; handling **document and media processing** and **WebSocket-based real-time communication**; secured **6 insurance clients**.

PricewaterhouseCoopers (PwC) **Aug 2021 – Aug 2024**
Senior Software Engineer
- Orchestrated a distributed backend on **Microsoft Azure**, leveraging **Azure Blob Storage** and event-driven architecture to process large datasets, reducing data processing time by **75%**.
- Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction, saving the engineering team **15 hours of manual work** per week.
- Led code reviews and mentored engineers on **software design principles**, improving code quality and reducing bugs by **25%**.
- Optimized data ingestion pipelines using **parallel processing** for **100K-row Excel datasets**, reducing execution time from **5 hours to 1 hour** and enabling daily refreshed analytics.
- Customized **Power BI dashboards** processing large datasets from Polyglot databases, improving reporting efficiency by **40%**.

Projects

Hospital Management System (Full-Stack Application) | ***React, Node.js, Express, MongoDB, SageMaker, Hugging Face***
- Built a **full-stack healthcare platform** using **React, Node.js, Express, and MongoDB** to manage appointments, patient records, lab reports, and medical history with **role-based access control**.
- Implemented a **graph-based doctor recommendation system** using **Dijkstra's algorithm** to match patients with the nearest available doctor, improving booking efficiency by **45%**.
- Fine-tuned a **Hugging Face transformer-based LLM** achieving **88% accuracy** for health condition prediction and deployed it on **AWS SageMaker** for real-time diagnostic support in the doctor dashboard.

Education

New York University | *Master of Science in Computer Engineering* ***(GPA – 3.9/4)*** Sep 2024 – May 2026

Certifications

AWS Cloud Practitioner
Microsoft Solution Architect Expert
Power BI Data Analyst Associate
Microsoft Developer Associate
Stanford Machine Learning (Coursera)
Azure Data Fundamentals